

MetaController: Sheet Material Based Flexible Game Controlling System

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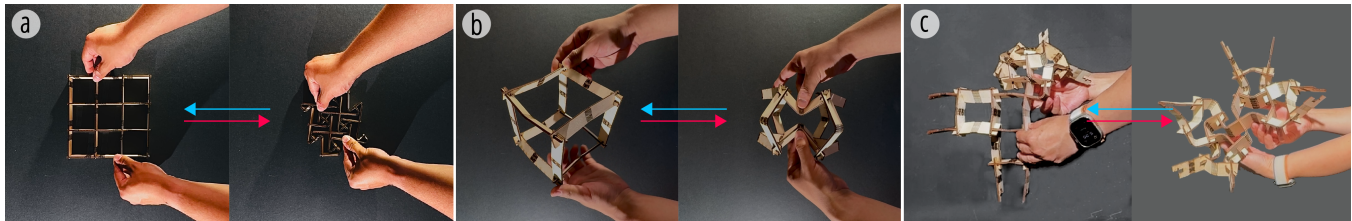


Figure 1: Three examples of assemblies using MetaController base units: (a) an input device that has auxetic property, (b) a space structure, (c) a freeform assembly whose limbs can be independently controlled.

Abstract

We introduce MetaControllers, reconfigurable game-controlling devices that are made of wood sheets. We intend to use this flexible material system to create devices whose configurations can be customized for different games to create context-specific haptic experiences, which cannot otherwise be achieved by traditional game controllers. Our approach exploits cellular-based material structure units consisting of rigid faces and hinges out of one single material. By assembling units together, we can create higher level of mechanical transformations through either embedded actuation or externally applied compression. In this proposal, we show that the different compositions of these units, when coupled with electronics, can enable novel ways of controlling and interacting with the games that provide dynamic physical affordances and guide users with dynamic physical constraints.

CCS Concepts

• Human-centered computing → Interaction design;

Keywords

Linkage Design, Digital Fabrication, Computational Design, Shape-changing Materials, Input and Output Technologies, Augmented Interactions

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1 Introduction and Related Work

The existing game controller design paradigm has largely homogenized the physical performance of game players. With a keyboard or joystick, the use of our bodies for playing Tetris is the same as playing Tic-Tac-Toe, or playing Flappy Birds. The constraining interaction modalities cause the loss of the richness, subtlety, and coordination of tasks at several levels, as well as hinder users' thinking and communication. Previous studies in psychology, sociology, and philosophy have shown that physical action and cognition are interconnected [10]. Gestures can largely lighten the cognitive load both in adults and children [7]. Therefore, devices that can provide “natural” mappings and leverage our familiarity with the real world can help to increase users' reasoning abilities and performance.

In the industry, recent trends have seen efforts from large gaming firms to shift the design paradigm of game controllers by providing more diverse and inclusive interactions in their products. For example, the Xbox Adaptive Controller allows players to create their own rig with the addition of custom buttons and joysticks based on the belief that gaming should be inclusive, accessible, and safe for all [1]. LABO kit from Nintendo that comes equipped with games that can be played by mapping the physical motions of the user to the in-game controls has been extremely well-received. In academia, researchers have made significant advancements in the design of haptic controllers for gaming experiences, resulting in various methods for prompting physical gestures and rendering sensations [3–5, 11]. However, the generalizability of the devices across games and demographic groups often contradicts with game-specific affordances, hence posing an engineering and design challenge. In our MetaController system, we focus on solving this conflict by proposing a reconfigurable material system that can be assembled and adapted to create various interactive modalities through different mechanical performances [9]. MetaControllers are made from wood sheets through a subtractive manufacturing

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process using affordable desktop laser cutters. With a simple and efficient fabrication and making workflow, users can design and tailor their own assemblies at home. Instead of conforming to the pre-designed mechanisms, users can make game controllers that best fit their physical conditions and preferences.

2 System-And-Implementation

2.1 Concept

MetaControllers are customizable and reconfigurable flexible material devices comprised of basic units, which enables the creation of monolithic affordable input devices for different gaming experiences. We accomplish this by laser cutting 3mm basswood panels for making basic units, coupling them with input and output technologies, and assemble them based on game-specific requirements. When a user interacts with the device and applies a force, measurable inputs, such as the distance and contacting area between opposing or neighboring cell walls, the vibration frequency and amplitude of the device, etc., change, allowing a new interaction design approach that better integrates the physical and computational worlds [7].

2.2 Flat Panels of a Single Material Engineered with Compliant Arrays for Emerging Motions

In MetaController system, we adopt compliant arrays to create tunable mechanical responses that are typically not achievable from a flat panel of a single material [8]. The 1-DOF wood hinges are engineered with slit-based patterns, analogous to the kinematic pairs in rigid-body mechanisms [2]. The removal of sections of the rigid material enables large amounts of elastic deflection. The ability to fabricate them from planar layers offers the potential for compact and lightweight devices that can be fabricated at low manufacturing cost [6].

2.3 Input and Output Technologies

We propose two pipelines for connecting the MetaController to the actual game/program.

For existing games, the microcontroller (XIAO ESP32C3) processes the data reading from input sensors and sends it to the serial port. Our program processes the data using pyautogui. The program projects the input to the keys used in the game and changes the game state.

For new games, developers can choose to have the microcontroller send the processed data directly to Unity as the serial port communication method is also supported by the program, therefore, providing possibilities for leveraging MetaController as an integral part of the design at the beginning of the process.

3 Use Case/User Scenario - Reconstructing interaction mode for Game Play

MetaController will use multiple sensors like Grove - Magnetic Switches (SKU 101020038), Grove - Rotary Angle Sensor (SKU 101020017), Grove - 3-Axis Digital Accelerometer (SKU 101020039), etc., which will allow it to support a variety of different games.

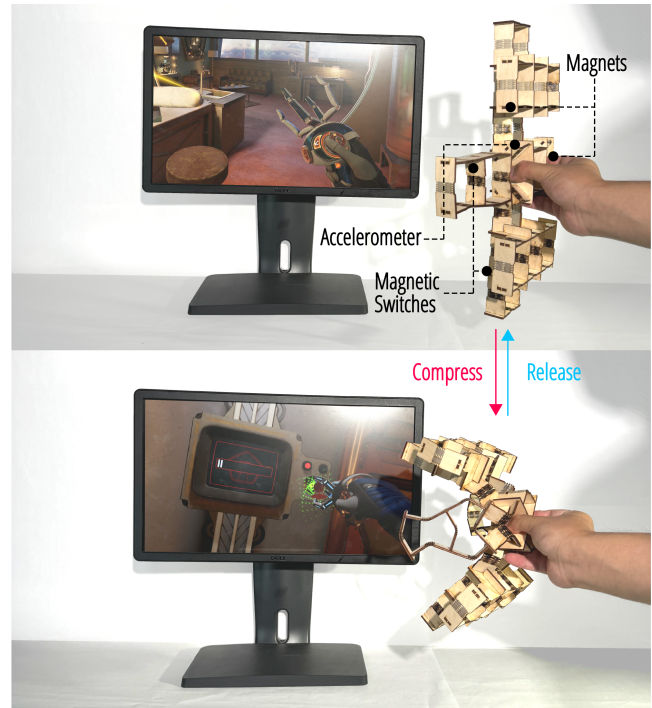


Figure 2: A freeform assembly of MetaController that features four limbs. The limbs can be controlled simultaneously by the central cell with compressive motion. The MetaController is coupled with a 3-axis Digital Accelerometer for detecting motions, two Magnetic Switches and Magnets for detecting the grip. The image illustrates a user using MetaController to play the Game *Red Matter 2*

Let's take an example of a game where the player character is controlling a claw to pick up objects and perform specific tasks. Traditionally, the player does that by pressing a button on a wireless controller. With our system, the player will combine the base units to form the Freeform Assembly structure as shown in Figure 2, which is augmented with a 3-axis digital accelerometer, two pairs of magnetic switches and magnets. When the player moves the controller structure, it creates a velocity change in the physical limbs. The accelerometer detects the change and sends the data to the microcontroller. When the player compresses the central cell, the limbs come closer. The shortened distance between the magnets and the switches triggers a signal change and sends it to the microcontroller. Adopting the key projection pipeline mentioned above, the physical input then can be used to coordinate the movement of the virtual limbs.

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